StyleReel

AMD DEVELOPER HACKATHON ACT II / TRACK 2

One video.
Four voices.

A containerized agent that captions any clip in four grounded voices, with the video understanding running on Gemma 3, self-hosted on AMD.

github.com/NishchayMahor/stylereel

THE TASK

Four voices, one clip, a 10 minute clock

Given short video clips, write a caption in four styles: formal, sarcastic, tech-humor, and everyday-humor. An LLM judge scores every caption on accuracy and tone.

Miss a style

and that clip scores zero.

Malformed JSON

and the whole batch scores zero.

Run past 10 minutes

and you score zero.

So the bar is not just good captions. It is good captions that **never fail to ship**.

THE REAL OUTPUT

Same footage, four ways to say it

FORMAL

A small orange kitten with large eyes walks forward through a sunlit forest, emerging from dense green foliage onto leaf-littered ground.

SARCASTIC

A kitten emerges from some foliage with the gravity of someone who has definitely remembered where they left something. Riveting.

TECH HUMOR

A kitten with intensely focused eyes steps out of the undergrowth. Classic on-call engineer, three steps into the incident runbook before the alert finished firing.

EVERYDAY HUMOR

A tiny kitten marches out of the bushes with the grim determination of someone who just remembered they left the stove on.

The facts stay true to the video. Only the tone changes.

UNDER THE HOOD

Architecture

SEE + HEAR	DESCRIBE	STYLIZE X4	JUDGE
Frames + audio	**Gemma 3 on AMD**	**Four writers**	**Keep the best**
ffmpeg smart frames, local Whisper transcript	watches the frames, writes one checked description	best of five, grounded only in the facts	independent model gates accuracy
	Radeon PRO W7900		

The describe stage runs on **GEMMA_ENDPOINT** with an automatic **Fireworks vision failover**, so the 10-minute batch always finishes. Every caption comes from one verified description, so the four never contradict the video or each other.

Gemma 3 12B | vLLM 0.16 + ROCm 7.2 | Fireworks AI | faster-whisper | Docker / linux/amd64

BEST USE OF GEMMA

Gemma watches the video, on AMD silicon

We did not just call Gemma over an API. We ran its multimodal vision ourselves: **Gemma 3 12B on an AMD Radeon PRO W7900** via vLLM and ROCm. It is the model that actually watches every clip.

> "a small orange tabby kitten with large eyes, walking forward, emerging from behind foliage, in a wooded area with sunlight filtering through the leaves creating dappled shadows."
>
> GEMMA 3 ON W7900, DESCRIBING THE CLIP

HEAD TO HEAD, SAME CLIPS AND JUDGE

DESCRIBE BACKEND	ACCURACY	STYLE
Gemma 3 on AMD W7900	**0.754**	**0.925**
Kimi vision (cloud)	0.763	0.925

Gemma **ties** our tuned cloud vision model (inside the judge's noise), with a simpler prompt and no verify pass. ~42 GB of the 48 GB VRAM resident; vision encoder profiled for 8 images.

WHY THE ACCURACY HOLDS

The understanding stage does the work

CONFIGURATION	ACCURACY
Full pipeline	**0.86**
No verification pass	0.87
No best-of-five	0.87
Blind, no describe stage	**0.72**

+0.15

Removing the describe stage drops accuracy from **0.86 to 0.72**. Understanding the video first, before any joke, is where the score is won. And that stage is exactly what Gemma owns.

BUILT TO NEVER SCORE ZERO

Resilience is a feature

Skeleton-first output

A valid results file exists from the first second and refreshes after every clip. A crash never leaves you with nothing.

Degraded ladder

Full pipeline, then skip judge, then skip best-of-N, then single-shot, then a grounded fallback. A style key is never missing.

Deadline watchdog

Results are written before any slow thread can push past the 10-minute kill. Then the process hard-exits.

Verified

5 of 5 chaos drills (dead URLs, odd styles, malformed input) still write valid JSON and exit clean. A 2-minute, 3-scene clip runs in 51 seconds.

MEASURED, NOT CLAIMED

Results

0.93	1.00	51s	5/5
STYLE MATCH	BLIND STYLE-ID	2-MIN, 3-SCENE CLIP	CHAOS DRILLS

RUN IT

```
# tasks.json in ./input, results.json in ./output
docker run --rm -v $PWD/input:/input:ro -v $PWD/output:/output ghcr.io/nishchaymahor/stylereel:latest
```

Repo: github.com/NishchayMahor/stylereel · Demo: nishchaymahor.github.io/stylereel